

MLB ParkCast: Weather & Play Across MLB Stadiums

DATASCI 192B, Capstone Project



Spring 2026

Ava Brown

Andrew
Christofferson

Colin McKhann

Rebecca Wu



Executive Summary

- Built MLB ParkCast: interactive dashboard that predicts park factor based on daily weather conditions and provides comprehensive analysis
- Major Historical Trends:
 - Temperature drives offense across the MLB
 - Pressure impacts stadiums the most uniformly (dome and open air)
 - Marine layer suppresses fly balls at West Coast night games
- Air density is most significant predictor of park factor
 - Temperature effect materializes through air density effect
- Wind Direction > Wind Speed

Detailed Review



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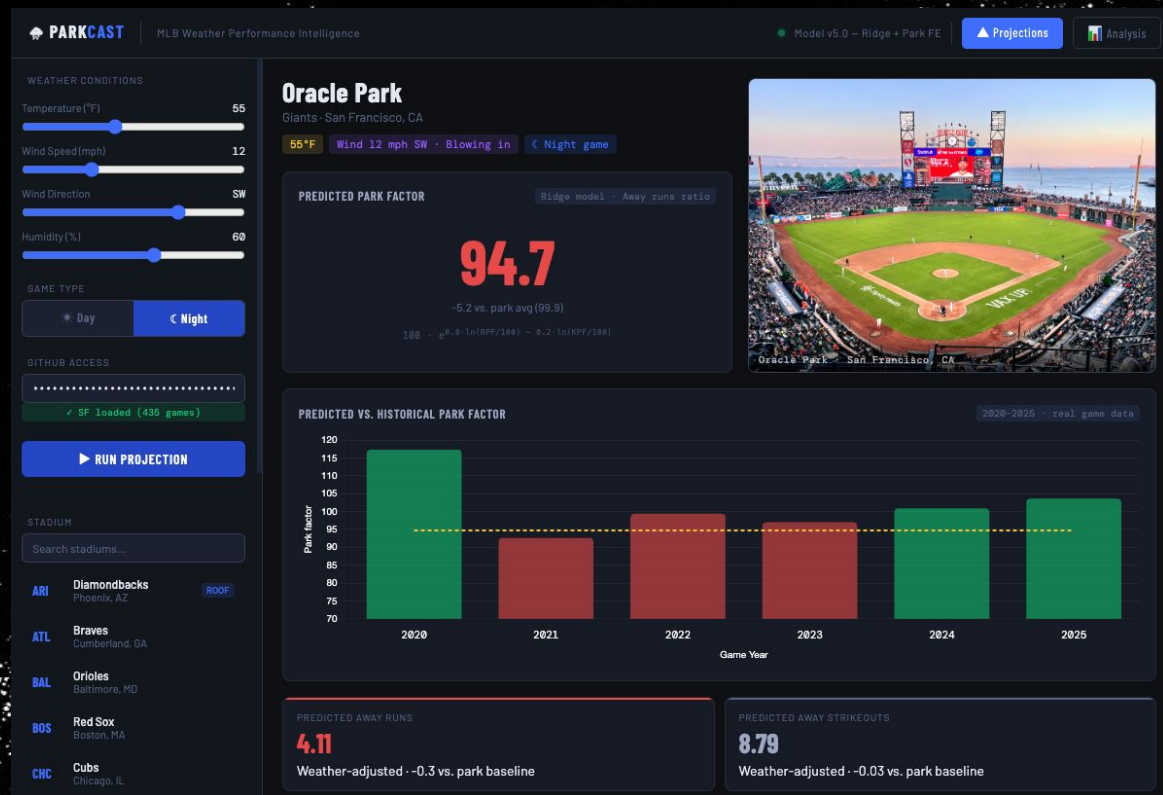
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Solution Overview



Existing Statistics

Point-in-time picture of historical weather and stadium impacts.

- Annual Park Factor
- Outfield Dimensions
- Batted-Ball Aggregations

MLB ParkCast

Dynamic prediction tool to analyze stadium specific effects on any given day.

- Comprehensive Stadium Analysis
- Historical Weather Patterns
- Real-time Predictions Based on Game Day Conditions



Literature Review

Petriello (2021) & Funf (2025)

Background information on the park factor statistic and how stadium characteristics influence the outcome of similar hits and pitches in different parks across the MLB.

Koch & Panorska (2013)

Relationship between temperature and baseball statistics. Key finding: home runs and runs scored increase with temperature.

Kagan & Mitchell (2017)

Marine layer effects account for over 90% of the variance in fly ball distance with an average effect of -6 feet. Relevant for coastal parks on the west coast (Oracle, Angel Stadium, etc).

Nunnally (2023)

Weather Applied Metrics (WAM) using fluid dynamics models are used to analyze on-field wind because environment wind varies drastically from conditions in the park.

Hunter (2014)

Comparison of fly ball/HR distance at the Rogers Centre with roof open vs closed → no convincing evidence that dome substantially changes likelihood of home runs.

Patterson (2025)

Broader analysis of weather metric effects (temperature, pressure, humidity, altitude) on fly balls. We used this research as a baseline comparison for our initial analysis.

Gerber & Craig (2021)

Using mixed effects multinomial logistic normal model to forecast baseball performance. Influenced our predictive model design (fixed effects and random effects terms).

Frost (2018)

While weather impacts baseball performance, it will never explain the entire game outcome. This article helped us interpret results in a lower r-squared setting.



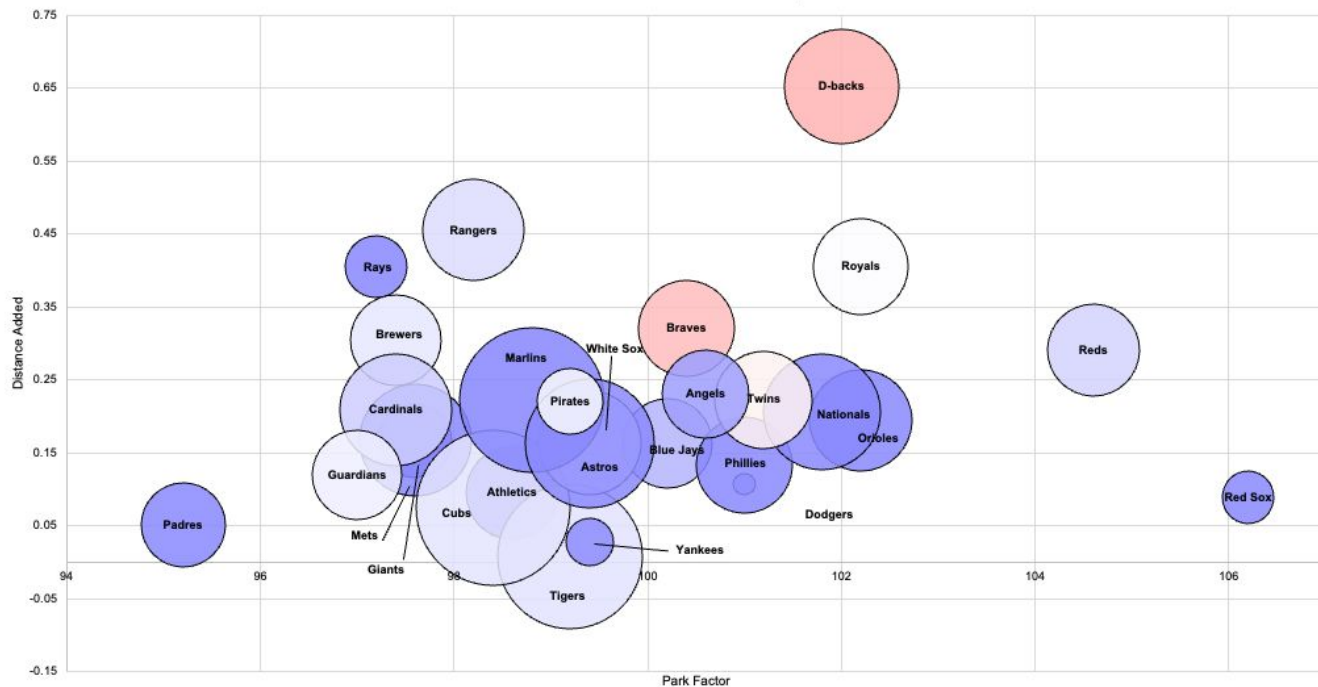
X-axis: 5-year rolling average of Park Factor

Y-axis: Index of net distance added due to environmental factors

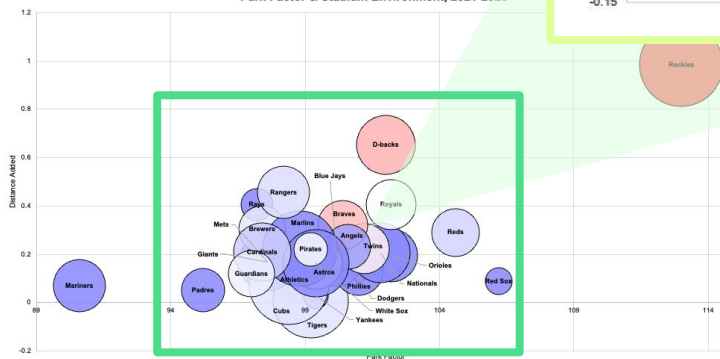
Bubble Size: Index of outfield size

Bubble Color: Elevation index (red = high elevation, blue = low elevation)

Park Factor & Stadium Environment, 2021-2025



Park Factor & Stadium Environment, 2021-2025



↑ Elevation, ↑ Park Factor

↑ Distance Added, ↑ Park Factor

Size and Park Factor more variable correlation



Methods

- ✓ Designing an interactive dashboard that goes beyond static annual park factors by using a predictive model to estimate daily park factor based on game day conditions and provides detailed information on historical weather patterns and park characteristics.

Data Cleaning

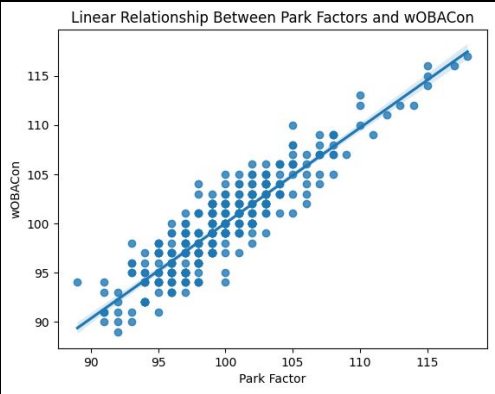
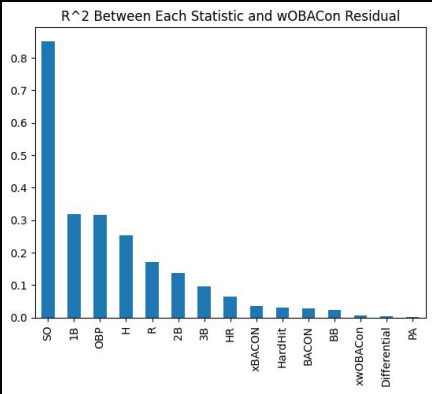
- Stadium-specific datasets
 - Environmental weather statistics averaged over the 3 hours following first pitch (OpenMeteo)
 - Aggregated pitch-level data to create game-level entries (Pybaseball Statcast, MLB Stats API)
 - Focused on away team performance to avoid bias of home team roster
- Feature Design: Wind
 - Project wind along vectors from home plate to LCF, CF, RCF (+ → blowing towards outfield)

Variable	Mean	Std	Min	Max
Game Start Hour	16.61	2.60	3.00	20.00
Away Home Runs	1.16	1.16	0.00	9.00
Away Runs Scored	4.43	3.24	0.00	28.00
Away Strikeouts	8.86	2.94	0.00	21.00
Away Walks	3.03	1.96	0.00	17.00
Temperature (°F)	71.69	12.36	20.50	112.30
Humidity (%)	63.16	19.68	2.30	100.00
Pressure (hPa)	996.19	31.37	824.10	1032.80
Precipitation (mm)	0.40	1.69	0.00	46.90
Wind Speed (mph)	7.73	3.87	0.50	35.90
Wind to CF	2.06	10.03	-38.01	43.52
Wind to LCF	1.92	10.20	-38.80	45.20
Wind to RCF	1.96	9.71	-39.47	43.96

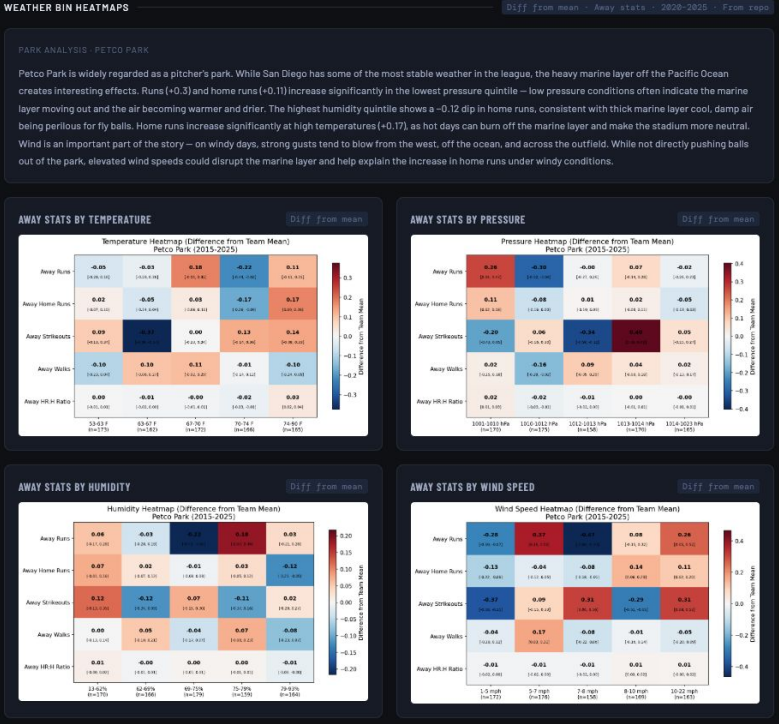
n = 12,058 games · Source: master_data.csv + league_weather_2021_2025.csv

Intro Data Analysis

- Define Park Factor for future modeling
- Analyze historical weather/play patterns across stadiums



$$RSKPF = 100 \times e^{w_R \ln(RPF/100) - w_K \ln(KPF/100)}$$



Heat map analysis → **count-level** impact of weather on game play with confidence interval + description

Modeling Approach

- Data: 2021-2025 game-level data by team + corresponding weather
- Features: temperature, humidity, wind speed, directional wind component (CF), air density
- Data Split: temporally by month (70/30)
- Model: mixed-effects regression
 - Fixed weather effects (average across parks)
 - Random intercepts: ball park, away team, season
 - Continuous features standardized
- 4% adjusted r-squared → weather should NOT explain all variability

Ridge Interaction Model Hierarchy

SF Giants Weather-Performance Analysis — Model v6.0.0

Level 1: Observation (Game)

y_{ijk} = Outcome for game i , in park j , by team k

Fixed Effects (Population-Level)

β_1 · temp_f Temperature	β_2 · rhum Humidity	β_3 · wspd_mph Wind Speed
β_4 · wind_cf Wind to CF	β_5 · air_density Air Density	β_6 · elev_ft Elevation
β_7 · is_night Night Game	β_8 · marine_day Marine Day	β_9 · marine_night Marine Night

Park × Weather Interactions (150 Terms)

γ_i · w_f for each park i and weather feature f

Park Intercepts

a_i for 30 parks
Baseline park effect

Park-Specific Slopes

γ_i (30 × 5 = 150)
How each park responds

Marine Layer

Day/Night effects
6 Pacific coast parks

Marine Layer:
SF, OAK, SD
LAK, LAD, SEA

Implicit Random Structure

Away Team Effect

$u_i \sim N(0, \sigma_{away}^2)$
30 MLB Teams

Season Effect

$v_k \sim N(0, \sigma_{season}^2)$
10 Seasons

Residual Error

$\epsilon_{ijk} \sim N(0, \sigma^2)$
Unexplained

Complete Model Equation (v6)

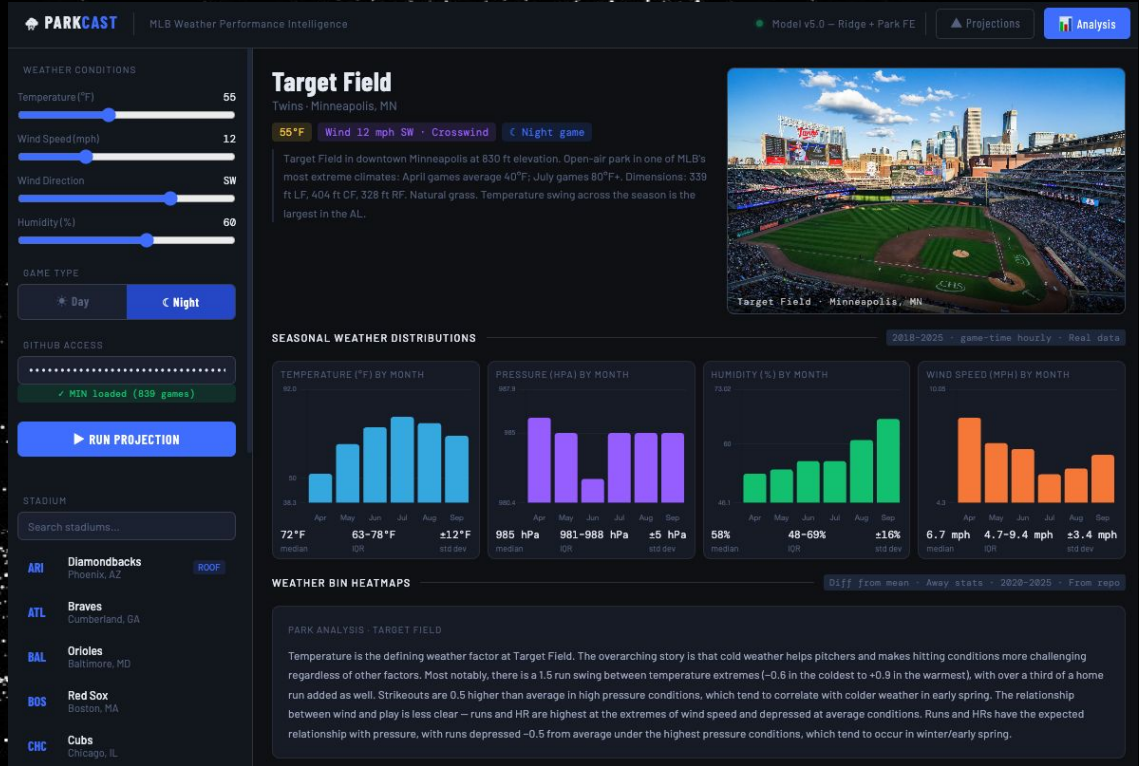
$$\hat{y}_i = \beta_0 + a_i + \sum_{f=1}^9 (\beta_f + \gamma_i) \cdot w_{if} + \beta_6 \cdot \text{elev} + \beta_7 \cdot \text{night} + \beta_8 \cdot \text{marine}$$

where: a_i = park intercept, γ_i = park-weather interaction, w_{if} = standardized weather, marine = day/night coastal effect

Dashboard Design

- Accessibility without sacrificing analytical depth
- Input panel: game day conditions
- Projections tab: predicted park factor and key determinants (inferential statistics)
- Analysis tab: historical park trends and detailed weather analysis by stadium (data exploration)

Interface



ParkCast Demo

Results

- ✓ Key takeaways and trends observed across the league through historical analysis and predictive modeling.

Predictive Findings: Key takeaways from the modeling process

1



League Wide Findings

Air density is the dominant predictor (2x any other variable)

- +0.41 strikeouts per 1 SD increase in air density
- -0.22 runs per 1 SD increase in density

Wind direction matters more than wind speed for run scoring

- Toward CF: +0.13 runs
- Speed alone: +0.07 runs

Temperature drives strikeouts (+0.17) but has near-zero direct effect on runs → works through air density

2



Park Specific Findings

1. Dodgers: wind toward CF increases K's (+0.30) and decreases runs (-0.25)
2. Cubs: Largest wind-to-run interaction in the MLB (+0.21)
3. Red Sox: Largest humidity-to-K's interaction in the MLB (+0.21)
4. A's: largest single weather interaction at +0.35 for humidity
5. Rockies: other variables have negative interactions because altitude maxes out effects
6. Twins: air density amplified at cold extremes (+0.14 Ks, -0.11 runs)

3



Oracle Insights

- Near neutral baseline (-0.07K, -0.01 runs)
- Wind suppresses more runs than average (-0.09 speed, -0.05 direction)
- The McCovey Cove winds blowing in compound league-wide effects.



Analytical Findings: League wide weather trends

1



Temperature Drives Offense

In open-air parks, colder temperatures consistently suppress runs and home runs while increasing strikeouts. Warmer games see the opposite effect.

2



Low Pressure Increases Runs

Low barometric pressure is the primary weather effect that impacts both domed and open air stadiums. Lower-pressure days produce more runs and home runs.

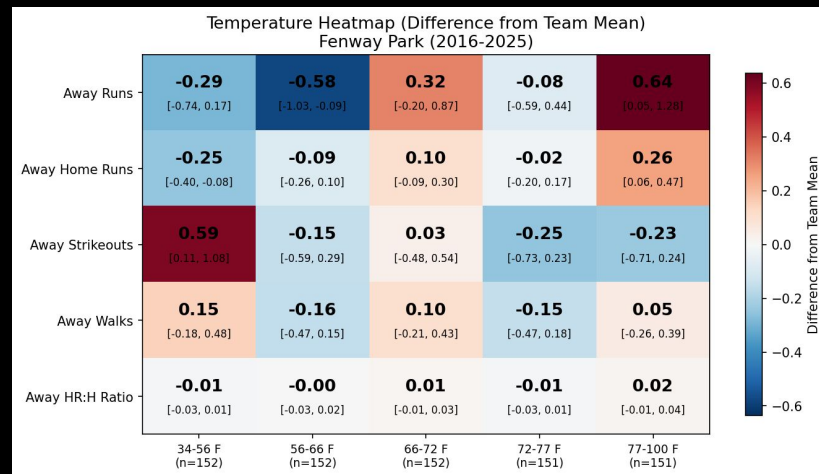
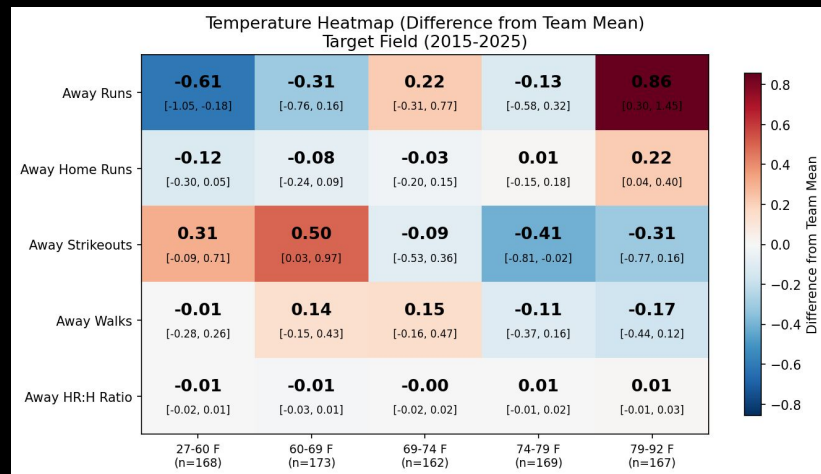
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Marine Layer Suppresses Carry

At coastal parks, the marine layer rolls in at night, bringing cooler temperatures, higher humidity, denser air, and stronger winds that suppress fly-ball carry.

1. Temperature & Offense in Open-Air Parks

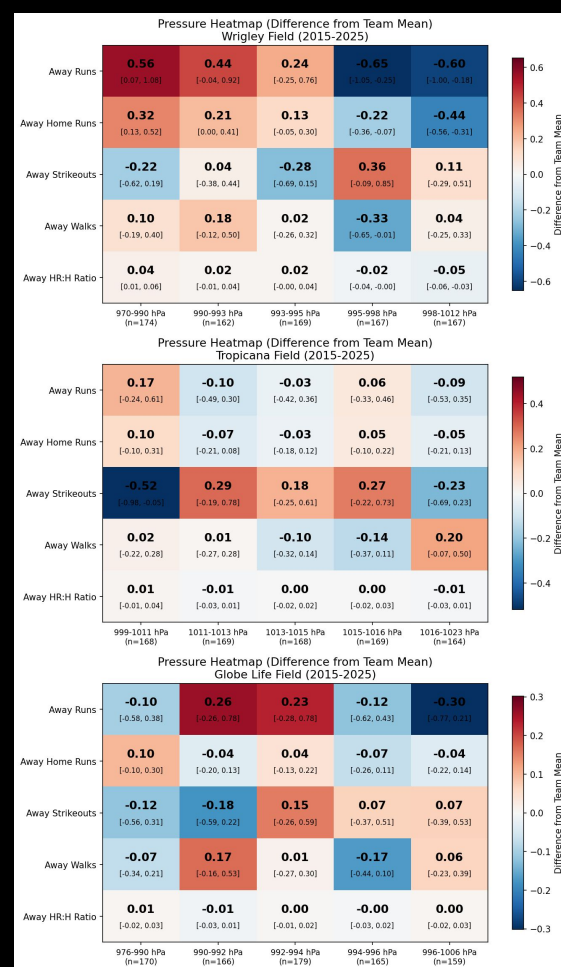


Temperature is the most consistent league-wide offensive driver in open-air parks and should be in every open-air park's park factor projection as a continuous variable. This pattern is directionally replicated across geographies at a more significant magnitude than most other effects. Northern parks see the largest swings because they have more significant cold-weather tails. For example, Target Field in Minnesota sees a 1.5 run swing between the 0-20th and 80-100th percentile. Boston mirrors this trend with a 0.95 run swing. Both parks also show strikeouts increasing in colder games and falling with temperature. We also see this trend at Camden Yards, Citizens Bank Park, Comerica Park, Kauffman Stadium, Wrigley Field, etc.

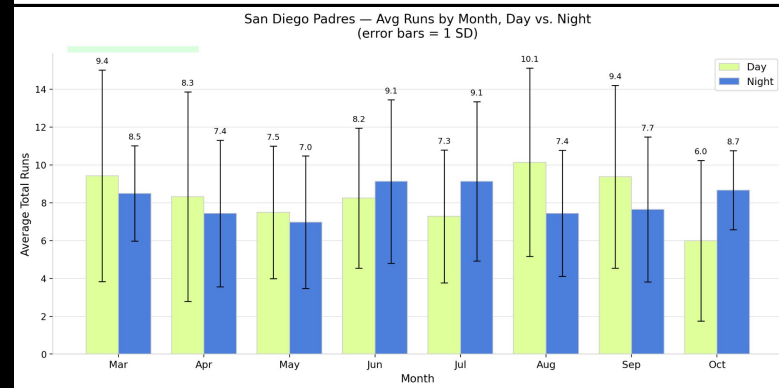
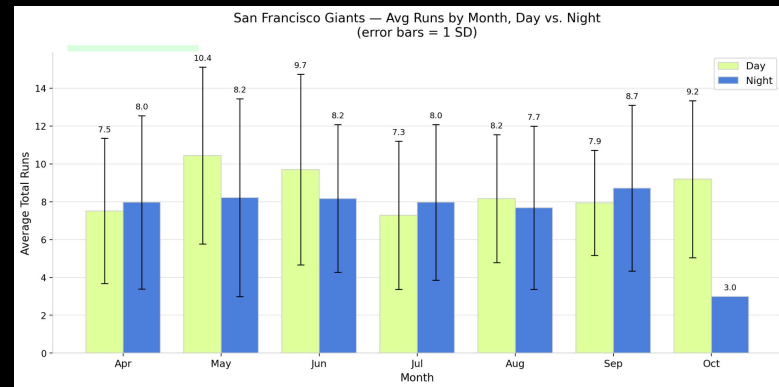
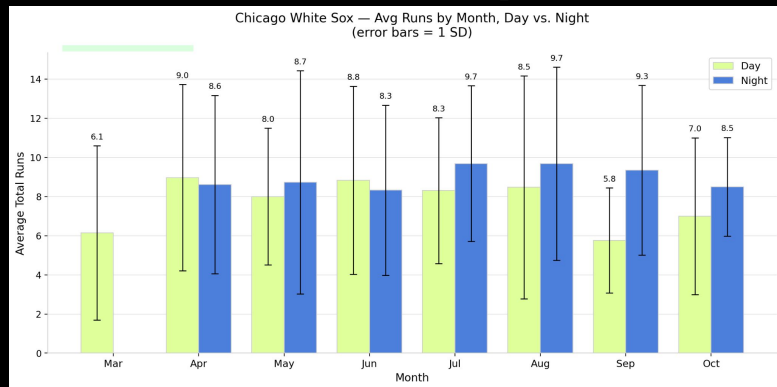
2. Pressure Matters (Even In a Dome)

Lower barometric pressure reduces aerodynamic drag on the ball, and the heatmap data shows this directly: across the league, low pressure days produce more runs and home runs, high pressure days produce more strikeouts. What makes this trend uniquely rigorous, is that it persists in enclosed and domed stadiums that are largely insulated from temperature, humidity, and wind. Pressure should be a prominent feature in a general, park-agnostic model because it continues to be impactful even in retractable and dome parks where most other weather input is noise.

Wrigley shows a clean open-air case with a 1.16 swing in runs and 0.76 swing in home runs across pressure quintiles. Tropicana Field is a fixed dome that still shows a negative correlation with pressure (although milder). Globe Life field has a retractable roof that is closed for over 80% of games and we see that runs typically increase on low pressure days and fall as pressure rises. These 3 stadiums show the strength of this trend across climates, geographies, and stadium types.



3. Marine Layer & Game Time



Runs drop significantly during night-time games at West Coast parks like Oracle (-0.54) and Petco (-0.52), but central and east coast parks do not see a similar effect (Rate Field +1.01, Fenway +0.32).

Day-vs-night time effects should be modeled as a coastal-park interaction, not as a league-wide fixed effect, and future iterations of the model should likely include a "marine-layer" indicator that will reflect the systematic offense suppression that basic temperature and humidity buckets will miss.

Next Steps

- ✓ Discussion of data and modeling limitations of the current approach and ways to build upon this research to develop more comprehensive tools to understand how weather changes play from game to game.

Limitations

Weather is not the primary determinant of a baseball game – how do you model effectively without all of predictive the inputs?

- Availability of in-stadium wind metrics
 - Accounted for via vectorizing wind direction, however can miss more specific potential interactions
- Lack of data for new stadiums
 - Model built off of past stadium data, so teams who have had recent major stadium adjustments (such as the Athletics relocating) have less useful input data
- Potential outlier impacts
 - Runs scored
 - Temperature
- Model readability → how do you assess efficacy when r-squared is logically low?

Areas for Continued Research

Marine Layer

Improve the marine layer flag for the impact of coastal conditions on day vs night games on the West Coast. Identify similar regional weather trends that would matter across multiple stadiums.

Wind Modeling

Include more advanced wind modeling for in stadium wind direction (would require private data). The behavior of wind inside the stadium is what is really driving play differences, and that cannot be fully captured by environmental data.

Player Aware Park Factor

Modify the park factor formula (which currently focuses on runs and strikeouts) to be aware of the players in the line-up on a given day and modify the park factor accordingly – we don't want to say a certain stadium is very offensive, when in reality it just hosted the best hitting team in the league.

Live Updates

Enhance the model to actively incorporate new game data throughout the season (perhaps with a heavier weighting) to make the daily park factor prediction more accurate and timely.

Thank you

Questions?

avabrown@stanford.edu
gcandrew@stanford.edu
colinmck@stanford.edu
rswu@stanford.edu

